Guidance and Control of Unmanned Autonomous Vehicles

(Course Code: AE 700)

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Project

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In this project, we will simulate some concepts covered during class. We will proceed systematically. Hence, the project goals have been presented in sections. The block diagram in Figure 1 shows an overview of the project. Each group needs to design the blocks named under "Part-1", by combining all the pieces of section-1 through section-5.

To get started:

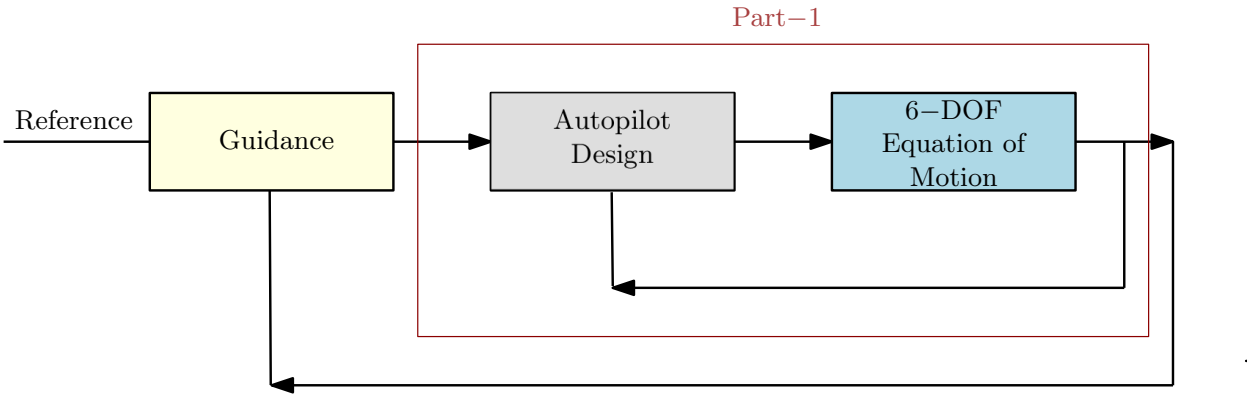

Figure 1: Schematic of the project

- Download the template code from the website.
- Pick the aircraft data file assigned to your group from this website.

Note: Assume any necessary missing data suitably and justify your assumptions.

1 Coordinate Frames

The objective is to create a 3-D graphic of an MAV that is correctly rotated and translated to the desired configuration. Creating animations in Simulink is described in Appendix C and example files are contained on the textbook website.

(i) Read Appendix C and study carefully the spacecraft animation using vertices and faces given on the textbook website.

(ii) Create an animation drawing of the aircraft.

(iii) Using a Simulink model like the one given on the website, verify that the aircraft is correctly rotated and translated in the animation.

(iv) In the animation file, switch the order of rotation and translation so that the aircraft is first translated and then rotated, and observe the effect.

2 Kinematics and Dynamics

Read Appendix D on building s–functions in Simulink, and also the Matlab documentation on s–functions.

(i) Implement the MAV equations of motion given in 3.14 through 3.17 using a Simulink s-function. Assume that the inputs to the block are the forces and moments applied to the MAV in the body frame. Block parameters should include the mass, the moments and products of inertia, and the initial conditions for each state. Use the parameters given in Aircraft Data. Simulink templates are provided on the website.

(ii) Connect the equations of motion to the animation block developed in the previous chapter. Verify that the equations of motion are correct by individually setting the forces and moments along each axis to a nonzero value and convincing yourself that the motion is appropriate.

(iii) Since J_{xz} is non-zero, there is gyroscopic coupling between roll and yaw. To test your simulation, set J_{xz} to zero and place nonzero moments on l and n, and verify that there is no coupling between the roll and yaw axes. Verify that when J_{xz} is not zero, there is coupling between the roll and yaw axes.

3 Forces and Moments

(i) Download the simulation files from the website. Modify the block `forces_moments.m`, which implements the gravity, aerodynamic, and propulsion forces and torques described in this chapter. Use the parameters given in Aircraft Data.

(ii) Modify the gust block to output wind gusts along the body axes. Modify `forces_moments.m` so that the outputs are the forces and moments resolved in the body frame, the airspeed V_a, the angle of attack α, the side-slip angle β, and the wind vectors resolved in the inertial frame $(w_n, w_e, w_d)^\top$.

(iii) Verify your simulation by setting the control surface deflections to different values. Observe the response of the MAV. Does it behave as you think it should?

4 Linear Design Models

(i) Read Appendix F and familiarize yourself with the Simulink `trim` and `linmod` commands.

(ii) Copy and rename your current Simulink diagram to `mavsim_trim.slx` and modify the file so that it has the proper input-output structure, as shown in Figure F.1 of Appendix F of [1].

(iii) Create a Matlab script that computes the trim values for the Simulink simulation developed in section 1 through 3. The input to the Matlab script should be the desired airspeed V_a, the desired path angle $\pm\gamma$, and the desired turn radius $\pm R$, where $+R$ indicates a right-hand turn and $-R$ indicates a left-hand turn.

(iv) Use the Matlab script to compute the trimmed state and controls for wings-level flight with $\alpha = 5^\circ - 7^\circ$ and $\gamma = 0$ rad. Set the initial states in your original Simulink simulation to the trim state, and the inputs to the trim controls. If the trim algorithm is correct, the MAV states will remain constant during the simulation. Run the trim algorithm for various values of γ. The only variable that should change is the altitude h. Convince yourself that the climb rate is correct.

(v) Use the Matlab script to compute the trimmed state and controls for constant turns with load factor, $n = 1.2$, and the lift coefficient, $C_L = 0.7 - 1.0$. Set the initial states in your original Simulink simulation to the trim state, and the inputs to the trim controls. If the trim algorithm is correct, the UAV states will remain constant during the simulation except for the heading ψ.

(vi) Create a Matlab script that uses the trim values computed in the previous problem to create the transfer functions listed in section 5.4 of the [1].

(vii) Create a Matlab script that uses the trim values and the `linmod` command to linearize the Simulink model about the trim condition to produce the state-space models given in equations 5.50 and 5.43 of the [1].

5 Autopilot design

The objective of this chapter is to design lateral and longitudinal autopilots for the assigned aircraft model. Please include the following plots in the report.

Lateral autopilot:

- Roll angle along with its commanded value vs. time without disturbance
- Roll angle along with its commanded value vs. time with disturbance
- Course angle, along with its commanded value vs. time
- Side slip angle along with its commanded value vs. time
- Control input signals for all these autopilots

Longitudinal autopilot

- Pitch angle, along with its commanded value vs. time
- Altitude, along with its commanded value vs. time
- Airspeed, along with its commanded value vs. time using throttle
- Airspeed, along with its commanded value vs. time using commanded pitch
- Control input signals for all these autopilots

Assume the bound on a surface deflection to be 40 degrees with suitably chosen maximum error in the tracking variable. You can use the assumption of slow variations in outputs for autopilot design. You need to submit the code for verification of your results of tracking in autopilot.

Note: Take the values of all the commanded values as step signals of suitable amplitude in the simulation. Also, give the disturbance as a step signal with a magnitude of 0.2.

Please ensure that the responses you attach to the report must be legible. Plots must include a label, legend, and title and have a suitable font size.

References

[1] Small Unmanned Aircraft: Theory and Practice Randal W. Beard and Timothy W. McLain